

CCNA Configuration & Troubleshooting Handbook

Practical CLI, Troubleshooting, and Best Practices Guide

By Charles Hendricks D. Solinap
Electronics Engineering Student | CCNA Candidate | Networking Enthusiast

Version 1.1 | 2026

TABLE OF CONTENTS

TABLE OF CONTENTS	i
Chapter 1: Basic Device & Security Management.....	1
Part 1: Navigation & CLI Shortcuts	1
Executing Exec-Level Commands from Global Config	1
Negating or Removing a Command	1
Disabling DNS Lookup	1
Filtering Output (The Pipe):	1
Part 2: Hostnames & Basic Passwords	1
Device Identification	1
Securing Privileged EXEC Mode	2
Part 3: Line Security (Console, VTY, SSH).....	2
Securing the Physical Console.....	2
Preparing for SSH (Remote Access).....	2
Securing the VTY Lines	2
Part 4: Global Password Encryption & Banners	3
Encrypting Plain-Text Passwords	3
Creating a Message of the Day (MOTD)	3
Part 5: Configuration Management	3
Saving the Configuration	3
Verification and Troubleshooting (Show Commands).....	3
Chapter 2: Layer 2 Switching & Spanning Tree.....	4
Part 1: Physical Ports & VLAN Creation	4
Basic Physical Port Configuration	4
VLAN Creation.....	4
Part 2: Access Ports & Voice VLANs	4
Part 3: Trunking, DTP, & VTP.....	4
Trunk Configuration	4
Dynamic Trunking Protocol (DTP)	5
VLAN Trunking Protocol (VTP)	5
Part 4: Spanning Tree Protocol (STP)	5
STP Modes & Root Bridge	5
Manipulating the Path	5
Part 5: STP Security Features.....	5
PortFast & BPDU Guard.....	5
Advanced STP Guards	6
Part 6: EtherChannel	6
Creating the Channel	6
Verification and Troubleshooting (Show Commands).....	6
Chapter 3: Layer 3 IP Routing.....	8
Part 1: Router Physical Interfaces & Loopbacks	8
Configuring a Physical Interface.....	8
Configuring a Loopback Interface	8
Part 2: IPv6 Addressing	8
Enabling IPv6 Routing.....	8
Assigning IPv6 Addresses	8
Enabling Link-Local Only	8
Part 3: Inter-VLAN Routing & L2 Switch Management	9
Method 1: Router-on-a-Stick (ROAS)	9

Method 2: Layer 3 Multilayer Switch	9
Method 3: Layer 2 Access Switch Default Gateway.....	9
Part 4: Static Routing (IPv4 & IPv6)	9
Standard Static Routes	9
Default Routes (Gateway of Last Resort).....	10
Floating Static Routes (Redundancy).....	10
Verification and Troubleshooting (Show Commands)	10
Chapter 4: Dynamic Routing (OSPFv2)	11
Part 1: OSPF Process & Router ID	11
Initializing the Process	11
Setting the Router ID	11
Part 2: Enabling OSPF on Interfaces (Two Methods)	11
Method 1: The Traditional Network Command	11
Method 2: The Interface-Level Command	11
Optimizing Point-to-Point Links:.....	12
Part 3: Passive Interfaces & Default Route Injection	12
Configuring Passive Interfaces.....	12
Injecting a Default Route	12
Part 4: Modifying OSPF Cost & Timers	12
Modifying the Reference Bandwidth	12
Manually Forcing Interface Cost.....	13
Modifying OSPF Timers	13
Verification and Troubleshooting (Show Commands)	13
Chapter 5: IP Services.....	14
Part 1: DHCPv4 Server & DHCP Relay	14
Configuring the DHCP Server.....	14
DHCP Relay (Helper Address).....	14
Part 2: Network Address Translation (NAT)	14
Prerequisite: Defining the Boundaries	14
Method 3: PAT / NAT Overload (Many-to-One)	15
Part 3: First Hop Redundancy Protocols (FHRP).....	15
Part 4: Network Time Protocol (NTP)	15
NTP Client Configuration	15
NTP Master Configuration	15
Verification and Troubleshooting (Show Commands)	15
Chapter 6: Network Security Fundamentals	17
Part 1: Switchport Port Security	17
Enabling Port Security	17
Configuring MAC Address Limits and Learning	17
Configuring Violation Modes.....	17
Part 2: DHCP Snooping & Dynamic ARP Inspection (DAI).....	17
DHCP Snooping.....	17
Dynamic ARP Inspection (DAI)	18
Part 3: IPv4 Access Control Lists (ACLs)	18
Standard ACLs (Filter by Source IP ONLY)	18
Extended ACLs (Filter by Source, Destination, Protocol, and Port)	18
Named ACLs	18
Part 4: Applying ACLs to Interfaces and VTY Lines.....	18
Applying to a Router Interface	18
Applying to VTY Lines (Securing SSH/Telnet)	18

Verification and Troubleshooting (Show Commands)	19
Chapter 7: Network Management, Operations, & Wireless	20
Part 1: Layer 2 Discovery (CDP & LLDP)	20
Cisco Discovery Protocol (CDP)	20
Link Layer Discovery Protocol (LLDP)	20
Part 2: Logging & Device Monitoring	20
Local Logging (Console, VTY, and Buffered)	20
Centralized Syslog	20
Simple Network Management Protocol (SNMP)	20
Part 3: File Management & IOS Upgrades	21
Backing up using TFTP, FTP, or SCP	21
Changing the Boot System	21
Part 4: Operations & System Recovery	21
Managing Hardware Tables (MAC & ARP)	21
Resetting an Interface	21
Wiping the Entire Device	21
Part 5: Wireless LAN Controllers (WLC Basics)	22
Connecting a Switch to a WLC (LAG)	22
Verification and Troubleshooting (Show Commands)	22

Chapter 1: Basic Device & Security Management

Before you can configure complex routing or switching, you must secure the device and understand how to navigate its operating system efficiently. These foundational commands are universally applied to almost every Cisco router and switch.

Part 1: Navigation & CLI Shortcuts

Efficiency on the CLI saves critical time during exams and real-world troubleshooting.

Executing Exec-Level Commands from Global Config

```
Router(config)# do [command]
```

Note: Normally, commands like show ip route or ping only work in Privileged EXEC mode (#). By prefixing them with do, you can run them from anywhere without having to exit your current configuration mode.

Negating or Removing a Command

```
Router(config)# no [command]
```

Best Practice: If you make a typo (like configuring the wrong IP address), always use no to remove the incorrect command before typing the correct one.

Disabling DNS Lookup

```
Router(config)# no ip domain-lookup
```

Note: By default, if you mistype a command in Privileged EXEC mode, the device assumes it is a hostname and attempts to resolve it via DNS, freezing the console for several seconds.

Best Practice: Apply this command immediately when setting up a new lab device to prevent frustrating delays caused by typos.

Filtering Output (The Pipe):

In a production environment, a core switch's configuration can be thousands of lines long. Scrolling through it is a massive waste of time. You can use the pipe character (|) to filter the output of any show command.

```
Router# show running-config | include [keyword]
```

```
Router# show running-config | section [keyword]
```

```
Router# show running-config | begin [keyword]
```

Note: include only shows the exact lines containing the keyword (e.g., ... | include ip address). section pulls out an entire related configuration block (e.g., ... | section ospf). begin skips straight to the first instance of the keyword and shows everything after it (e.g., ... | begin line vty).

Best Practice: Master the pipe | before your OJT. Using it during a technical interview or on the job is one of the fastest ways to prove you are comfortable navigating a live Cisco IOS device.

Part 2: Hostnames & Basic Passwords

Securing the transition from User EXEC mode (>) to Privileged EXEC mode (#).

Device Identification

```
Router(config)# hostname [name]
```

Note: A hostname is strictly required before you can generate RSA keys for SSH access.

Securing Privileged EXEC Mode

```
Router(config)# enable password [password]
Router(config)# enable secret [password]
Router(config)# enable algorithm-type scrypt secret [password]
Router(config)# username [username] algorithm-type scrypt secret [password]
```

Note: enable password stores the password in plain text. enable secret uses an older hash (MD5). The scrypt algorithm uses a highly secure Type 9 hash.

Note: If both are configured simultaneously, the device will prioritize enable secret and ignore enable password.

Best Practice: Never use enable password. Always use algorithm-type scrypt if the IOS version supports it. It is significantly more secure against brute-force attacks. If the device is older, fall back to standard enable secret.

Part 3: Line Security (Console, VTY, SSH)

Securing both physical (Console) and remote (VTY) access to the device.

Securing the Physical Console

```
Router(config)# line console 0
Router(config-line)# password [password]
Router(config-line)# login
Router(config-line)# logging synchronous
Router(config-line)# exec-timeout [minutes] [seconds]
```

Note: logging synchronous prevents syslog messages (like an interface going down) from interrupting the command you are currently typing.

Best Practice: Always configure an exec-timeout (e.g., exec-timeout 5 0 for 5 minutes) to automatically log out unattended console sessions.

Preparing for SSH (Remote Access)

To use SSH instead of the insecure Telnet protocol, you must complete these prerequisites first.

```
Router(config)# ip domain-name [domain-name]
Router(config)# crypto key generate rsa
Router(config)# username [username] secret [password]
```

Note: When generating the RSA key, you will be prompted for a modulus size. Use at least 1024 for SSHv2, though 2048 is the modern standard.

Securing the VTY Lines

```
Router(config)# line vty 0 15
Router(config-line)# login local
Router(config-line)# transport input ssh
```

Note: login local tells the line to authenticate users against the local database you created with the username command, rather than a single shared line password.

Best Practice: Using transport input ssh explicitly disables Telnet, forcing all remote management sessions to be encrypted.

Part 4: Global Password Encryption & Banners

Encrypting Plain-Text Passwords

`Router(config)# service password-encryption`

Note: This applies a weak "Type 7" Cisco-proprietary encryption to any plain-text passwords residing in the configuration file (like console or vty passwords).

Note: This is only meant to prevent "shoulder surfing" and can be easily reversed with online decrypters.

Creating a Message of the Day (MOTD)

`Router(config)# banner motd [delimiter] [message] [delimiter]`

Note: The delimiter is any character that does not appear in the message itself (commonly # or %). For example: banner motd # Unauthorized Access is Prohibited #.

Part 5: Configuration Management

Changes made in the CLI are stored in RAM (the running-config) and will be permanently lost if the device loses power.

Saving the Configuration

`Router# copy running-config startup-config`

Note: This copies your active RAM configuration into NVRAM (Non-Volatile RAM), ensuring the device loads it upon the next reboot.

Note: You may also see write memory or just wr used in production environments. While copy run start is the official standard for the CCNA, wr is a deeply ingrained habit for many engineers and does the exact same thing on IOS devices.

Verification and Troubleshooting (Show Commands)

Use these commands from Privileged EXEC mode (#) to verify Chapter 1 configurations.

`show running-config`

- **Usage:** Displays the entire active configuration currently running in RAM.
- **Troubleshooting:** Use this to verify your passwords are encrypted (Type 7 or Secret hashes), that your hostnames match your topology, and that your interfaces are assigned the correct IP addresses.

`show startup-config`

- **Usage:** Displays the saved configuration residing in NVRAM.
- **Troubleshooting:** If this returns "startup-config is not present", you have not saved your work yet. If a device is acting strangely, you can compare the running and startup configs to see what recent changes might be causing the issue.

`show version`

- **Usage:** Displays system hardware, software version, uptime, and the Configuration Register.
- **Troubleshooting:** If your device is booting up blank despite you saving the configuration, check the Configuration Register at the very bottom of this output. 0x2102 is normal; 0x2142 tells the router to bypass NVRAM on boot (used for password recovery).

`show ip ssh`

- **Usage:** Verifies that SSH is enabled and confirms the version running.

`show users`

- **Usage:** Displays all active console and VTY sessions, including the source IP address of remote users.

Chapter 2: Layer 2 Switching & Spanning Tree

Part 1: Physical Ports & VLAN Creation

Before assigning ports, the physical interfaces should be configured, and the VLANs must exist in the switch's local database.

Basic Physical Port Configuration

```
Switch(config)# interface range [interface-id] - [interface-id]
Switch(config-if-range)# speed [10 | 100 | 1000 | auto]
Switch(config-if-range)# duplex [half | full | auto]
Switch(config-if-range)# description [description-text]
```

Best Practice: Leave speed and duplex set to auto on modern networks to avoid duplex mismatches, which cause severe packet loss. Use the interface range command to configure multiple interfaces simultaneously to save time.

VLAN Creation

```
Switch(config)# vlan [vlan-id]
Switch(config-vlan)# name [vlan-name]
```

Best Practice: Always apply a descriptive name to your VLANs. Naming conventions are critical in a real-world environment to quickly identify traffic types (e.g., HR_DEPT or GUEST_WIFI) during an outage.

Part 2: Access Ports & Voice VLANs

Access ports belong to a single data VLAN and usually connect to end hosts like PCs or printers.

```
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan [vlan-id]
Switch(config-if)# switchport voice vlan [vlan-id]
```

Note: The switchport voice vlan command tells the switch to use an 802.1Q tag for the voice traffic (for an IP phone) while leaving the connected PC's data traffic untagged.

Best Practice: Never leave an end-device port in its default dynamic state. Hardcoding it with switchport mode access permanently disables trunk negotiation, mitigating VLAN hopping attacks.

Part 3: Trunking, DTP, & VTP

Trunk ports carry traffic for multiple VLANs across a single physical link, typically between two switches.

Trunk Configuration

```
Switch(config-if)# switchport trunk encapsulation [dot1q | isl]
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk native vlan [vlan-id]
Switch(config-if)# switchport trunk allowed vlan [add | remove | all | none | except] [vlan-list]
```

Note: encapsulation dot1q is required on older multi-layer switches. Newer switches default to 802.1Q and will reject the command. ISL is Cisco proprietary and obsolete.

Best Practice: Always configure the native VLAN to an unused, dummy VLAN ID. Use the add keyword when modifying allowed VLANs to avoid accidentally overwriting the entire list and dropping all production traffic.

Dynamic Trunking Protocol (DTP)

```
Switch(config-if)# switchport mode dynamic [auto | desirable]
Switch(config-if)# switchport nonegotiate
```

Note: DTP is Cisco proprietary.

Best Practice: DTP is a security risk. Disable it by statically configuring your trunks (switchport mode trunk) and applying switchport nonegotiate to stop the transmission of DTP frames.

VLAN Trunking Protocol (VTP)

```
Switch(config)# vtp domain [domain-name]
Switch(config)# vtp mode [server | client | transparent]
Switch(config)# vtp version [1 | 2 | 3]
```

Note: VTP is Cisco proprietary.

Best Practice: Set VTP to transparent mode to manually manage VLANs locally. This prevents a rogue switch with a higher configuration revision number from accidentally wiping your entire network's VLAN database.

Part 4: Spanning Tree Protocol (STP)

STP prevents Layer 2 loops by blocking redundant links. Cisco switches run Per-VLAN Spanning Tree (PVST) by default.

STP Modes & Root Bridge

```
Switch(config)# spanning-tree mode [pvst | rapid-pvst | mst]
Switch(config)# spanning-tree vlan [vlan-list] root [primary | secondary]
Switch(config)# spanning-tree vlan [vlan-list] priority [priority-value]
```

Note: rapid-pvst is the Cisco proprietary implementation of the faster 802.1w standard. The priority-value must be configured in increments of 4096.

Best Practice: Always use rapid-pvst for fast convergence (1-3 seconds instead of 30-50). Manually enforce your Root Bridge placement using the root primary or priority commands on your core distribution switches so edge switches don't take over the topology.

Manipulating the Path

```
Switch(config-if)# spanning-tree vlan [vlan-list] cost [cost-value]
Switch(config-if)# spanning-tree vlan [vlan-list] port-priority [priority-value]
```

Note: Port-priority is configured in increments of 32. Lower cost and lower port-priority are preferred by STP when determining the best path to the Root Bridge.

Part 5: STP Security Features

These features protect the STP topology from unexpected changes or unauthorized devices.

PortFast & BPDU Guard

```
Switch(config-if)# spanning-tree portfast
Switch(config-if)# spanning-tree bpduguard enable
Switch(config)# errdisable recovery cause bpduguard
Switch(config)# errdisable recovery interval [seconds]
```

Note: If BPDU Guard receives a BPDU on a PortFast-enabled port, it forces the port into an err-disabled state. The default recovery interval is 300 seconds if configured globally.

Best Practice: ONLY enable PortFast on access ports facing end devices. Always pair it with BPDU Guard to protect against users plugging unauthorized switches into wall jacks.

Advanced STP Guards

```
Switch(config-if)# spanning-tree guard root
```

```
Switch(config-if)# spanning-tree guard loop
```

Note: Only one between root guard and loop guard can be enabled on an interface.

Best Practice: Apply guard root on all downstream ports facing access switches to prevent them from becoming the Root Bridge. Apply guard loop on redundant uplinks to prevent loops caused by unidirectional fiber link failures.

Part 6: EtherChannel

EtherChannel bundles multiple physical links into one logical connection to increase bandwidth and provide redundancy without STP blocking the extra links.

Creating the Channel

```
Switch(config-if)# channel-group [group-number] mode [active | passive |
desirable | auto | on]
```

```
Switch(config)# interface port-channel [group-number]
```

```
Switch(config)# port-channel load-balance [src-mac | dst-mac | src-ip | dst-ip |
src-dst-ip]
```

Note: Modes active/passive use LACP (Industry Standard). Modes desirable/auto use PAgP (Cisco Proprietary). Mode on forces the channel without negotiation.

Note: Once the channel is created, you must apply your trunking/allowed VLAN configurations to the logical interface port-channel, not the physical ports.

Best Practice: Always use LACP (active) to ensure cross-vendor compatibility and to verify link health before bundling. Change the load-balance method from the default (src-mac) to src-dst-ip for better traffic distribution across the physical wires.

Verification and Troubleshooting (Show Commands)

Use these commands from Privileged EXEC mode (#) to verify Chapter 2 configurations.

```
show interfaces status
```

- **Usage:** Quick check of physical port assignments, connected status, speed, and duplex.

```
show vlan brief
```

- **Usage:** Verifies VLAN existence and displays which access ports are assigned to which VLANs.

```
show interfaces trunk
```

- **Usage:** Verifies active trunk links, Native VLANs, and the allowed VLAN list.
- **Troubleshooting:** If an expected VLAN is missing from the "allowed and active" list, it was likely improperly pruned.

```
show spanning-tree
```

- **Usage:** The ultimate command to view the STP topology. Identifies the Root Bridge, local Bridge ID, and port roles (Root, Designated, Alternate, Blocking).

```
show etherchannel summary
```

- **Usage:** Verifies that the logical Port-Channel is "SU" (Layer 2, In Use) and that physical ports are "P" (Bundled in port-channel).
- **Troubleshooting:** If ports say "s" (Suspended) or "I" (Stand-alone), check for configuration mismatches (speed, allowed VLANs, duplex) between the member ports.

Chapter 3: Layer 3 IP Routing

Part 1: Router Physical Interfaces & Loopbacks

Unlike switchports, router interfaces are pure Layer 3 boundaries. They require an IP address to function and are administratively disabled by default.

Configuring a Physical Interface

```
Router(config)# interface [interface-id]
Router(config-if)# description [description-text]
Router(config-if)# ip address [ip-address] [subnet-mask]
Router(config-if)# no shutdown
```

Note: The no shutdown command is strictly required to bring the port out of the administratively down state.

Best Practice: Always apply a description detailing the connected device or subnet. In an enterprise environment, undocumented router links make troubleshooting extremely difficult.

Configuring a Loopback Interface

```
Router(config)# interface loopback [interface-number]
Router(config-if)# ip address [ip-address] [subnet-mask]
```

Note: Loopback interfaces are logical, virtual interfaces. They are immediately brought "Up/Up" upon creation and never go down unless manually shut down or the router itself loses power.

Best Practice: Always configure a loopback interface on your routers. They provide a highly stable IP address that is perfect for use as a Router ID in protocols like OSPF.

Part 2: IPv6 Addressing

IPv6 routing is disabled by default on Cisco routers. You must explicitly enable the global process before the router will forward IPv6 packets.

Enabling IPv6 Routing

```
Router(config)# ipv6 unicast-routing
```

Note: If you forget this command, you can still assign IPv6 addresses to interfaces, but the router will act like a host and will not route traffic between subnets.

Assigning IPv6 Addresses

```
Router(config-if)# ipv6 address [ipv6-address]/[prefix-length]
Router(config-if)# ipv6 address [ipv6-prefix]/[prefix-length] eui-64
Router(config-if)# ipv6 address autoconfig
Router(config-if)# ipv6 address [ipv6-address]/[prefix-length] anycast
```

Note: eui-64 calculates the host portion of the address automatically using the interface's MAC address, injecting FFFE in the middle.

Note: autoconfig uses Stateless Address Autoconfiguration (SLAAC) to dynamically learn the network prefix from a neighboring router.

Note: anycast configures an address shared by multiple routers, providing redundancy for services.

Enabling Link-Local Only

```
Router(config-if)# ipv6 enable
```

Note: This automatically generates an FE80:: link-local address without requiring a global unicast address to be assigned.

Part 3: Inter-VLAN Routing & L2 Switch Management

To route traffic between different VLANs, a Layer 3 device must intervene. However, pure Layer 2 switches still need a gateway to communicate outside their local management subnet.

Method 1: Router-on-a-Stick (ROAS)

A single physical router interface connects to a switch trunk port and uses logical subinterfaces to route between VLANs.

```
Router(config)# interface [interface-id].[subinterface-number]
Router(config-subif)# encapsulation dot1q [vlan-number]
Router(config-subif)# ip address [ip-address] [subnet-mask]
```

Note: You must configure the encapsulation dot1q command to define the VLAN before the IOS will allow you to assign an IP address.

Best Practice: Match the subinterface number to the VLAN ID for easier management (e.g., interface g0/0.10 for VLAN 10). To configure the native VLAN on a ROAS setup, append the keyword: encapsulation dot1q [vlan-number] native.

Method 2: Layer 3 Multilayer Switch

Modern enterprise networks use Layer 3 switches to handle routing at wire speeds.

```
Switch(config)# ip routing
Switch(config)# interface [interface-id]
Switch(config-if)# no switchport
Switch(config-if)# ip address [ip-address] [subnet-mask]
```

Note: ip routing globally enables Layer 3 capabilities on the switch.

Note: no switchport strips the Layer 2 switching logic from a physical port, turning it into a routed port that accepts an IP address directly, exactly like a standard router.

Method 3: Layer 2 Access Switch Default Gateway

If a pure Layer 2 switch needs to send Syslog messages to a server in a different VLAN or reply to pings from an admin subnet, it must know where to send that traffic.

```
Switch(config)# ip default-gateway [gateway-ip]
```

Best Practice: Apply this globally on all Layer 2 access switches, pointing it to the HSRP Virtual IP (VIP) of their respective management VLAN.

Part 4: Static Routing (IPv4 & IPv6)

Static routes are manually entered into the routing table. They have an Administrative Distance (AD) of 1.

Standard Static Routes

```
Router(config)# ip route [dest-network-ip] [subnet-mask] [next-hop-ip]
Router(config)# ip route [dest-network-ip] [subnet-mask] [exit-interface]
```

Note: Using a [next-hop-ip] requires the router to perform a recursive lookup (finding how to reach the next hop). Using the [exit-interface] directly points the traffic out a specific port.

Best Practice: On multi-access Ethernet networks, always use the [next-hop-ip]. Using an exit interface on Ethernet causes the router to send ARP requests for every destination IP, exhausting resources. Only use the [exit-interface] on point-to-point serial links.

Default Routes (Gateway of Last Resort)

```
Router(config)# ip route 0.0.0.0 0.0.0.0 [next-hop-ip]
Router(config)# ipv6 route ::/0 [next-hop-ipv6]
```

Note: A default route matches all traffic that does not have a more specific entry in the routing table.

Floating Static Routes (Redundancy)

```
Router(config)# ip route [dest-network-ip] [subnet-mask] [next-hop-ip]
[administrative-distance]
Router(config)# ipv6 route [dest-network-ip6]/[prefix-length] [exit-interface]
[next-hop-ipv6] [administrative-distance]
```

Note: By changing the AD (e.g., to 115), you create a backup route. It will sit invisibly in the configuration and only inject itself into the routing table if the primary route (like an OSPF route with an AD of 110) fails.

Note: The IPv6 example above is a "fully-specified" static route (using both the exit interface and the next-hop IP). This is often required for IPv6 floating routes over Ethernet to ensure the router knows exactly how to resolve the backup path.

Verification and Troubleshooting (Show Commands)

Use these commands from Privileged EXEC mode (#) to verify Chapter 3 configurations.

`show ip interface brief / show ipv6 interface brief`

- **Usage:** Your first stop for troubleshooting. Verifies IP address assignments and interface status.
- **Troubleshooting:** If the Status is "Administratively Down", apply no shutdown. If the Status is "Up" but Protocol is "Down", you have a Layer 2 issue (e.g., a clock rate missing on a serial link, an encapsulation mismatch, or the connected device is down).

`show ip route / show ipv6 route`

- **Usage:** Displays the global routing table.
- **Troubleshooting:** Look for the legend codes. C is Connected, L is Local host, S is Static. If you configure a static route but it does not appear in this table, the exit interface it relies on is likely physically down.

`ping [ip-address]`

- **Usage:** Tests end-to-end Layer 3 reachability using ICMP.

`traceroute [ip-address] (Router) / tracert [ip-address] (Windows)`

- **Usage:** Details every router hop the packet takes to reach the destination.
- **Troubleshooting:** If a ping fails, use traceroute to find exactly which router in the path is dropping the packet, then check that specific router's routing table.

Chapter 4: Dynamic Routing (OSPFv2)

OSPF is a Link-State routing protocol. It builds a complete topographical map of the network and uses Dijkstra's Shortest Path First (SPF) algorithm to find the best routes. It has an Administrative Distance (AD) of 110.

Part 1: OSPF Process & Router ID

To start OSPF, you must initialize the routing process and give the router a name (Router ID) so other routers can identify it.

Initializing the Process

```
Router(config)# router ospf [process-id]
```

Note: The [process-id] is a locally significant number (1 to 65535). It does not have to match the neighboring routers to form an adjacency.

Setting the Router ID

```
Router(config-router)# router-id [ipv4-address]
```

```
Router(config-router)# clear ip ospf process
```

Note: The Router ID looks like an IP address (e.g., 1.1.1.1), but it is just a 32-bit name. If you do not manually configure one, OSPF will automatically choose the highest IP address on a Loopback interface, or the highest IP address on an active physical interface.

Note: If you change the Router ID after OSPF has already started, you must return to Privileged EXEC mode (#) and use clear ip ospf process to reset the protocol and apply the new name.

Best Practice: Always manually hardcode the Router ID. If you rely on physical interfaces, and that interface goes down, the Router ID changes, forcing the entire network to recalculate its SPF tree, which causes temporary network drops.

Part 2: Enabling OSPF on Interfaces (Two Methods)

You must tell OSPF which interfaces should participate in the routing process.

Method 1: The Traditional Network Command

This is configured inside the OSPF router configuration mode.

```
Router(config)# router ospf [process-id]
```

```
Router(config-router)# network [network-ip] [wildcard-mask] area [area-id]
```

Note: A wildcard mask is the inverse of a subnet mask (e.g., a /24 subnet mask of 255.255.255.0 has a wildcard mask of 0.0.0.255).

Note: The CCNA strictly focuses on Single-Area OSPF. You will almost always use area 0 (the backbone area).

Method 2: The Interface-Level Command

This is configured directly on the physical interface itself.

```
Router(config)# interface [interface-id]
```

```
Router(config-if)# ip ospf [process-id] area [area-id]
```

Best Practice: Use Method 2. It is the modern standard. Calculating wildcard masks leaves room for human error. Activating OSPF directly on the interface is exact, cleaner to read in the running-config, and explicitly shows which links are participating.

Optimizing Point-to-Point Links:

By default, OSPF on Ethernet interfaces assumes a multi-access network (like a switch with multiple routers) and wastes time electing a Designated Router (DR) and Backup Designated Router (BDR).

```
Router(config-if)# ip ospf network point-to-point
```

Best Practice: When connecting two routers or multilayer switches directly to each other (usually via a /30 subnet), hardcode the network type to point-to-point on both interfaces. This completely bypasses the unnecessary DR/BDR election and dramatically speeds up neighbor convergence.

Part 3: Passive Interfaces & Default Route Injection

Not all active OSPF interfaces should send OSPF Hello packets. If a port connects to a switch full of PCs, sending routing updates is a security risk and wastes bandwidth.

Configuring Passive Interfaces

```
Router(config-router)# passive-interface [interface-id]
```

```
Router(config-router)# passive-interface default
```

```
Router(config-router)# no passive-interface [interface-id]
```

Note: A passive interface still advertises its connected subnet to the rest of the OSPF network, but it stops sending out OSPF Hello messages, preventing rogue devices from forming neighbor adjacencies.

Best Practice: In a secure enterprise network, use passive-interface default to silence OSPF on all ports, and then selectively use no passive-interface only on the specific uplinks connected to other trusted routers.

Injecting a Default Route

```
Router(config-router)# default-information originate
```

```
Router(config-router)# default-information originate always
```

Note: If your edge router has a static default route pointing to the ISP (ip route 0.0.0.0 0.0.0.0 [isp-ip]), this command tells OSPF to advertise that default route to all internal routers automatically.

Note: The always keyword forces the router to advertise the default route even if its own static default route goes down.

Part 4: Modifying OSPF Cost & Timers

OSPF uses "Cost" as its metric. The cost is calculated by dividing a Reference Bandwidth by the interface's actual bandwidth.

Modifying the Reference Bandwidth

```
Router(config-router)# auto-cost reference-bandwidth [megabits-per-second]
```

Note: The default reference bandwidth is 100 Mbps. This means OSPF calculates FastEthernet (100 Mbps), GigabitEthernet (1000 Mbps), and 10-Gigabit as having the exact same cost (a cost of 1).

Best Practice: Always change the reference bandwidth to match the fastest link in your entire network (e.g., auto-cost reference-bandwidth 10000 for a 10Gbps backbone). You must apply this identical command to every single OSPF router in the topology for the routing math to be accurate.

Manually Forcing Interface Cost

```
Router(config-if)# ip ospf cost [cost-value]
```

Note: Instead of letting OSPF calculate the math, you can manually hardcode a cost directly on the interface to manipulate traffic flow and force it to take a specific path.

Modifying OSPF Timers

```
Router(config-if)# ip ospf hello-interval [seconds]
```

```
Router(config-if)# ip ospf dead-interval [seconds]
```

Note: By default, Hello packets are sent every 10 seconds, and the Dead timer is 40 seconds.

Troubleshooting: Hello and Dead timers must match exactly between two connecting routers. If they do not match, the neighbor adjacency will immediately drop.

Verification and Troubleshooting (Show Commands)

OSPF troubleshooting is heavily tested on the CCNA. Use these Privileged EXEC (#) commands to find out why your routers aren't talking.

```
show ip ospf neighbor
```

- **Usage:** Your primary OSPF diagnostic tool. It lists all connected OSPF routers.
- **Troubleshooting:** Look at the "State" column. You want to see FULL (or FULL/DR, FULL/BDR). If the state is stuck in INIT or EXSTART, you likely have an MTU mismatch or an access-list blocking OSPF traffic. If a router isn't listed here at all, check for mismatched Hello timers, mismatched Area IDs, or passive interfaces blocking Hello packets.

```
show ip ospf interface brief
```

- **Usage:** Displays a clean table of all interfaces running OSPF, their assigned Area, Cost, and their state (DR, BDR, or DROTHER).
- **Troubleshooting:** If an interface is missing from this list, you forgot to apply the network command or the ip ospf interface command.

```
show ip route ospf
```

- **Usage:** Filters the global routing table to show only the networks learned via OSPF (indicated by the O code).

```
show ip ospf
```

- **Usage:** Displays global OSPF process information, including the local Router ID, the number of times the SPF algorithm has been executed, and the current Reference Bandwidth.

Chapter 5: IP Services

Part 1: DHCPv4 Server & DHCP Relay

Instead of manually configuring IP addresses on hundreds of PCs, routers can act as DHCP servers to dynamically lease out IP configurations.

Configuring the DHCP Server

```
Router(config)# ip dhcp excluded-address [low-ip] [high-ip]
Router(config)# ip dhcp excluded-address [single-ip]
Router(config)# ip dhcp pool [pool-name]
Router(dhcp-config)# network [network-ip] [subnet-mask]
Router(dhcp-config)# default-router [gateway-ip]
Router(dhcp-config)# dns-server [dns-ip]
Router(dhcp-config)# lease [days] [hours] [minutes]
```

Note: The lease command is optional. The default lease time on a Cisco IOS device is 1 day.

Best Practice: Always exclude addresses (like the router's own interface IP, printer IPs, and server IPs) before creating the DHCP pool. If you don't, the router might assign an IP that is already in use, causing an IP conflict that drops a user off the network.

DHCP Relay (Helper Address)

Broadcasts (like DHCP Discover messages) are strictly stopped by routers. If your DHCP server is on a different network than your PCs, you must configure the router interface facing the PCs to forward those broadcasts as unicast traffic.

```
Router(config-if)# ip helper-address [dhcp-server-ip]
```

Part 2: Network Address Translation (NAT)

Private IPv4 addresses cannot be routed on the public internet. NAT translates your internal private IPs into routable public IPs provided by your ISP.

Prerequisite: Defining the Boundaries

You must explicitly tell the router which interface faces the internal network and which faces the internet. This is the #1 reason NAT configurations fail.

```
Router(config-if)# ip nat inside
Router(config-if)# ip nat outside
```

Method 1: Static NAT (One-to-One)

Maps a single private IP permanently to a single public IP. Typically used for internal web servers that need to be accessible from the outside.

```
Router(config)# ip nat inside source static [local-private-ip] [global-public-ip]
```

Method 2: Dynamic NAT (Many-to-Many)

Translates internal hosts to a pool of public IPs.

```
Router(config)# access-list [acl-number] permit [network-ip] [wildcard-mask]
Router(config)# ip nat pool [pool-name] [start-public-ip] [end-public-ip] netmask [subnet-mask]
Router(config)# ip nat inside source list [acl-number] pool [pool-name]
```

Method 3: PAT / NAT Overload (Many-to-One)

Port Address Translation maps hundreds of internal private IPs to a single public IP (usually the router's outside interface) by tracking unique port numbers. This is the standard deployment for almost all networks.

```
Router(config)# access-list [acl-number] permit [network-ip] [wildcard-mask]
Router(config)# ip nat inside source list [acl-number] interface [exit-interface]
overload
```

Best Practice: Always remember the overload keyword at the very end of the command. Without it, the router will only translate one internal host and then drop everyone else's traffic.

Part 3: First Hop Redundancy Protocols (FHRP)

If a network's only default gateway router dies, the entire subnet loses connectivity. Hot Standby Router Protocol (HSRP) links two physical routers together to share a single "Virtual IP" and "Virtual MAC," providing seamless failover.

Note: HSRP is Cisco proprietary. The open-standard equivalent is VRRP.

```
Router(config-if)# standby version 2
Router(config-if)# standby [group-number] ip [virtual-ip-address]
Router(config-if)# standby [group-number] priority [priority-value]
Router(config-if)# standby [group-number] preempt
```

Note: The default priority is 100. The router with the highest priority becomes the "Active" router.

Note: standby version 2 supports IPv6 and millisecond timers. Version 1 is the legacy default.

Best Practice: Always configure preempt on your primary router. If the primary router loses power and a backup takes over, the backup will keep the Active role forever. preempt forces the primary router to take its job back once it boots up again.

Part 4: Network Time Protocol (NTP)

If an outage occurs at 3:00 AM, but your router's internal clock says it's 1993, correlating Syslog messages to find the root cause is impossible.

NTP Client Configuration

```
Router(config)# clock timezone [timezone-name] [offset-from-utc]
Router(config)# ntp server [ntp-server-ip]
```

NTP Master Configuration

If you want a core router to act as the time source for all other internal switches.

```
Router(config)# ntp master [stratum-level]
```

Note: Stratum level determines the "distance" from the authoritative time source (1-15). Lower is better.

Verification and Troubleshooting (Show Commands)

Use these commands from Privileged EXEC mode (#) to verify Chapter 5 configurations.

```
show ip dhcp binding
```

- **Usage:** Displays all active MAC-to-IP-address leases currently handed out by the router.

```
show ip dhcp pool
```

- **Usage:** Shows pool statistics, including total addresses, leased addresses, and excluded address ranges.

```
show ip nat translations
```

- **Usage:** Displays the active NAT table.
- **Troubleshooting:** If you try to ping the internet and this table is completely empty, your ACL is likely configured incorrectly and isn't matching the internal traffic.

`show ip nat statistics`

- **Usage:** Confirms which interfaces are assigned as inside and outside, and shows the number of successful translations (hits) or failures (misses).

`show standby brief`

- **Usage:** The ultimate HSRP verification command. Shows the group number, priority, state (Active or Standby), the active router's IP, and the shared Virtual IP.

`show ntp status`

- **Usage:** Verifies if the router is synchronized with an NTP server and displays its current stratum level.

Chapter 6: Network Security Fundamentals

Part 1: Switchport Port Security

Port Security restricts a switchport so that only specific, authorized MAC addresses can send traffic through it, preventing attackers from plugging in unapproved switches or running MAC flooding attacks.

Enabling Port Security

```
Switch(config-if)# switchport mode access
Switch(config-if)# switchport port-security
```

Note: Port security cannot be enabled on a dynamic port. You must hardcode the port as an access port (or a trunk port, though rare for port security) first.

Configuring MAC Address Limits and Learning

```
Switch(config-if)# switchport port-security maximum [number]
Switch(config-if)# switchport port-security mac-address [mac-address]
Switch(config-if)# switchport port-security mac-address sticky
```

Note: sticky tells the switch to dynamically learn the first MAC address it sees and permanently save it to the running configuration.

Best Practice: Use a maximum of 1 or 2 (if an IP phone is attached) and use sticky learning so you don't have to manually type in hundreds of MAC addresses across the enterprise.

Configuring Violation Modes

```
Switch(config-if)# switchport port-security violation [protect | restrict | shutdown]
```

Note: protect drops unauthorized frames silently. restrict drops them and generates a Syslog message/SNMP trap. shutdown (the default) puts the port into an err-disabled state and sends a log message.

Best Practice: Leave the violation mode as shutdown for maximum security. If a port goes into err-disable, you must manually issue shutdown followed by no shutdown to recover it, which forces the administrator to investigate the security breach.

Part 2: DHCP Snooping & Dynamic ARP Inspection (DAI)

These two features work together to prevent Man-in-the-Middle (MitM) attacks at Layer 2.

DHCP Snooping

Prevents attackers from connecting a rogue DHCP server and handing out malicious IP configurations.

```
Switch(config)# ip dhcp snooping
Switch(config)# ip dhcp snooping vlan [vlan-list]
Switch(config-if)# ip dhcp snooping trust
Switch(config-if)# ip dhcp snooping limit rate [packets-per-second]
Switch(config)# no ip dhcp snooping information option
```

Note: By default, all ports are untrusted and will drop incoming DHCP Server replies. You must apply the trust command on your uplink ports facing your legitimate DHCP server.

Troubleshooting: By default, a switch running DHCP Snooping injects "Option 82" (circuit ID data) into the client's DHCP request. Some DHCP servers (especially Windows servers or older IOS routers) will drop the request because they don't understand the option. If your snooping config is correct but clients aren't getting IPs, use no ip dhcp snooping information option to stop the switch from altering the packet.

Best Practice: Apply rate limiting on untrusted access ports to prevent an attacker from starving your legitimate DHCP server with millions of fake DHCP Discover requests.

Dynamic ARP Inspection (DAI)

Relies on the DHCP Snooping binding database to inspect ARP packets. If an attacker tries to send a fake ARP reply (ARP Poisoning), DAI drops it.

```
Switch(config)# ip arp inspection vlan [vlan-list]
Switch(config-if)# ip arp inspection trust
Switch(config)# ip arp inspection validate src-mac dst-mac ip
```

Best Practice: Just like DHCP Snooping, all ports are untrusted by default. Configure trust on your uplinks. Enable the validate command globally to force the switch to do an advanced check of the MAC and IP addresses inside the ARP packet payload, providing maximum security.

Part 3: IPv4 Access Control Lists (ACLs)

ACLs are a set of rules used to filter traffic or identify traffic for other features (like NAT).

Standard ACLs (Filter by Source IP ONLY)

```
Router(config)# access-list [1-99] [permit | deny] [source-ip] [wildcard-mask]
Router(config)# access-list [1-99] [permit | deny] host [specific-ip]
Router(config)# access-list [1-99] [permit | deny] any
```

Extended ACLs (Filter by Source, Destination, Protocol, and Port)

```
Router(config)# access-list [100-199] [permit | deny] [protocol] [src-ip] [src-wildcard] [dst-ip] [dst-wildcard] eq [port-number/name]
```

Note: Common protocols include ip, tcp, udp, icmp, ospf. Common ports include 80 (www), 443 (443), 22 (ssh).

Named ACLs

```
Router(config)# ip access-list [standard | extended] [acl-name]
Router(config-ext-nacl)# [sequence-number] [permit | deny] [rule-parameters]
```

Best Practice: Always use Named ACLs in modern networks. They allow you to easily delete a specific line (e.g., no 20) or insert a new rule between existing ones by specifying a sequence number (e.g., 15 permit...). Numbered ACLs require you to delete the entire list and start over to make changes.

Part 4: Applying ACLs to Interfaces and VTY Lines

An ACL does absolutely nothing until it is applied to an interface or a line.

Note: Every ACL has an invisible "implicit deny all" at the very bottom. If traffic does not match any of your permit statements, it will be dropped.

Applying to a Router Interface

```
Router(config-if)# ip access-group [acl-number | acl-name] [in | out]
```

Best Practice: Standard ACLs should be applied as close to the destination as possible (because they only check source IPs and might accidentally block valid traffic if placed too soon). Extended ACLs should be applied as close to the source as possible (to drop unwanted traffic before it wastes bandwidth traversing the network).

Applying to VTY Lines (Securing SSH/Telnet)

```
Router(config)# line vty 0 15
```

Router(config-line)# access-class [acl-number | acl-name] in

Best Practice: Always secure your VTY lines with a Standard ACL that only permits management traffic from your specific IT Administrator subnet.

Verification and Troubleshooting (Show Commands)

show port-security interface [interface-id]

- **Usage:** Displays the port security status, violation mode, maximum MAC addresses, and the current security violation count for a specific interface.

Troubleshooting: If a port is err-disabled, check the violation count here to confirm if port security was the trigger.

show ip dhcp snooping

- **Usage:** Verifies which VLANs have snooping enabled and lists which interfaces are configured as trusted.

show ip arp inspection interfaces

Usage: Verifies which interfaces are trusted for DAI and shows the current ARP packet rate limits.

show access-lists

- **Usage:** Displays all standard, extended, and named ACLs configured on the router, along with the number of times each specific rule has been "hit" (matched) by traffic.

Troubleshooting: If a permit rule has 0 matches, but the implicit deny at the bottom is dropping traffic, your ACL rule is written incorrectly or applied in the wrong direction.

show ip interface [interface-id]

- **Usage:** A massive output command, but crucial for finding the "Outgoing access list is..." and "Inbound access list is..." lines to verify if your ACL was actually applied to the port.

Chapter 7: Network Management, Operations, & Wireless

Part 1: Layer 2 Discovery (CDP & LLDP)

Discovery protocols allow Cisco devices to share information about themselves (hostname, IP, port connected, IOS version) with directly connected neighbors.

Cisco Discovery Protocol (CDP)

```
Router(config)# cdp run
Router(config-if)# cdp enable
Router(config-if)# no cdp enable
```

Note: CDP is Cisco proprietary and is enabled globally and on all interfaces by default.

Link Layer Discovery Protocol (LLDP)

```
Router(config)# lldp run
Router(config-if)# lldp transmit
Router(config-if)# lldp receive
```

Note: LLDP is the IEEE 802.1AB open standard. Unlike CDP, it is usually disabled by default. You have granular control to allow an interface to receive LLDP packets but not transmit them.

Best Practice: Both protocols are a massive security risk if left running on ports facing end-users or the internet. Always use no cdp enable and disable LLDP on all untrusted access ports.

Part 2: Logging & Device Monitoring

Managing how a device reports its health and events is critical for incident response.

Local Logging (Console, VTY, and Buffered)

Instead of relying only on a live console session, you can carve out a piece of the router's RAM to store log messages so you can read them later.

```
Router(config)# logging console [severity-level]
Router(config)# logging buffered [size-in-bytes] [severity-level]
Router(config)# terminal monitor
```

Note: terminal monitor is an EXEC-level command (#) used when you are logged in remotely via SSH. Without it, you will not see live logs appear on your remote screen.

Best Practice: Always configure a buffered log of at least 16000 bytes. If an issue happens overnight, the buffered log is your only evidence of what went wrong.

Centralized Syslog

```
Router(config)# logging [syslog-server-ip]
Router(config)# logging trap [0-7]
Router(config)# logging source-interface [interface-id]
Router(config)# service timestamps log datetime msec
```

Note: The trap level determines severity (0 is Emergencies, 7 is Debugging). Level 6 (Informational) is common for capturing interface state changes.

Best Practice: Always configure service timestamps and a source-interface (usually a Loopback). Without timestamps, your logs are useless for correlation.

Simple Network Management Protocol (SNMP)

SNMP allows a central server to poll the router for CPU usage, bandwidth utilization, and health metrics.

```
Router(config)# snmp-server community [string-password] [ro | rw]
```

Note: This is SNMPv2c. ro is Read-Only, rw is Read-Write. SNMPv3 is the modern standard because it requires username authentication and encrypts the payload.

Part 3: File Management & IOS Upgrades

Routers and switches need their operating systems (IOS) and configuration files backed up regularly.

Backing up using TFTP, FTP, or SCP

```
Router# copy running-config tftp:
Router# copy running-config ftp:
Router# copy running-config scp:
```

Note: You will be prompted for the remote host IP and the destination filename. For FTP, you must first configure ip ftp username [name] and ip ftp password [password]. For SCP, you must configure ip scp server enable.

Best Practice: Always use SCP over FTP or TFTP if the network supports it. SCP uses SSH to securely encrypt the file transfer.

Changing the Boot System

If you load a new IOS image into flash memory, you must tell the router to boot from it.

```
Router(config)# boot system flash:[filename.bin]
```

Best Practice: After entering the boot system command, you MUST save the configuration (copy run start). If you just reload the router without saving, it will ignore your command and boot the old operating system.

Part 4: Operations & System Recovery

Commands used to force a network convergence or wipe a misconfigured device.

Managing Hardware Tables (MAC & ARP)

```
Switch# clear mac address-table dynamic
Router# clear arp-cache
Router# clear ipv6 neighbors
```

Troubleshooting: If you swap out a server's NIC, its IP remains the same but its MAC address changes. The router will send traffic to the old MAC until the ARP cache times out. Using clear arp-cache forces the router to immediately find the new MAC.

Resetting an Interface

```
Router(config)# default interface [interface-id]
```

Note: Instantly wipes all configuration under that specific interface, returning it to its factory-default state.

Wiping the Entire Device

```
Router# erase startup-config
Router# delete vlan.dat
Router# reload
```

Note: On switches, the VLAN database is often stored in a separate flash file called vlan.dat. If you don't delete it, the old VLANs will survive the wipe.

Part 5: Wireless LAN Controllers (WLC Basics)

While WLC configurations are mostly done via GUI, you must know how to configure the switchport that connects to the physical appliance.

Connecting a Switch to a WLC (LAG)

```
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# switchport mode trunk
Switch(config-if)# channel-group [group-number] mode on
```

Note: Unlike standard switch-to-switch EtherChannels that use LACP (mode active), connections to a WLC typically require the EtherChannel to be hardcoded to mode on with no negotiation protocol.

Verification and Troubleshooting (Show Commands)

`show cdp neighbors / show lldp neighbors`

- **Usage:** Displays directly connected devices, the local interface used, the remote hostname, and its platform type. Append detail to see the neighbor's management IP address.

`show logging`

- **Usage:** Displays the IP of the configured Syslog server, the current logging level, and the local buffer of the most recent log messages.

`show boot`

- **Usage:** Verifies the current boot variable string to confirm the router is pointed to the correct .bin file before a reload.

`show mac address-table / show ip arp`

- **Usage:** Displays the dynamically learned Layer 2 and Layer 3 forwarding tables.

`show version`

- **Usage:** Displays the currently running IOS version, uptime, and the Configuration Register. Check the Configuration Register at the bottom (0x2102 is normal; 0x2142 bypasses NVRAM for password recovery).